

Year 8 Science – Booklet 7: Chemistry

Reactivity Series — ANSWERS

Quick Test (p.65)

1. By electrolysis.
2. By observing how vigorously each metal reacts with air (oxygen), water, and dilute acid.
3. A metal oxide.
4. Zinc oxide.
5. $2\text{Zn} + \text{O}_2 \rightarrow 2\text{ZnO}$
6. Potassium hydroxide and hydrogen.
7. $2\text{K} + 2\text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{H}_2$

Section A – Multiple Choice (5 marks)

1. c) magnesium oxide
2. a) magnesium
3. d) metal hydroxide and hydrogen
4. c) calcium hydroxide and hydrogen
5. a) hydrogen

Section B (20 marks)

1. Complete the word equations

- a) magnesium + oxygen $\rightarrow$ magnesium oxide
- b) iron + oxygen $\rightarrow$ iron oxide
- c) copper + oxygen $\rightarrow$ copper oxide
- d) potassium + water $\rightarrow$ potassium hydroxide + hydrogen
- e) sodium + water $\rightarrow$ sodium hydroxide + hydrogen
- f) calcium + water $\rightarrow$ calcium hydroxide + hydrogen
- g) magnesium + hydrochloric acid $\rightarrow$ magnesium chloride + hydrogen
- h) magnesium + nitric acid $\rightarrow$ magnesium nitrate + hydrogen
- i) calcium + hydrochloric acid $\rightarrow$ calcium chloride + hydrogen
- j) zinc + hydrochloric acid $\rightarrow$ zinc chloride + hydrogen

2. Complete the symbol equations

- a) $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$
- b) $2\text{Fe} + \text{O}_2 \rightarrow 2\text{FeO}$
- c) $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$

Section C (8 marks)

1. Teacher's table (silver, sodium, magnesium, zinc)

- a). Most reactive $\rightarrow$ least reactive: **Sodium, Magnesium, Zinc, Silver.**
- b). Hydrogen.
- c). Magnesium chloride.

d) i). A metal not in the table that reacts like silver (no reaction): gold (or another very unreactive metal, e.g. platinum).

d) ii). A metal not in the table that reacts like sodium (vigorous with water): potassium (or calcium/lithium).

2. Calcium + hydrochloric acid $\rightarrow$ calcium chloride + hydrogen

a). Calcium (or hydrogen) – the two elements in the equation; hydrochloric acid and calcium chloride are compounds.

b). A metal that reacts more vigorously than calcium: sodium (or potassium).

c). A metal that reacts less vigorously than calcium: magnesium (or zinc, iron, lead, copper).